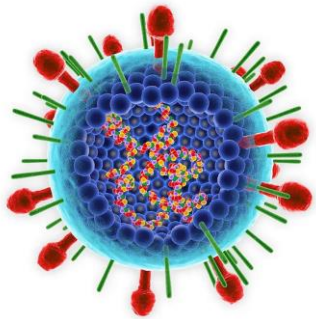


The Belgian National Survey on RSV pediatric burden of disease

Preliminary results

April 2025



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and the Belgian Pediatric RSV Study Group and the
BELSARI-net research group



Setting



- ▶ **New preventive tools against RSV disease** in children available from 2024

Monoclonal antibodies (Beyfortus®)/Maternal vaccine (Abrysvo®)

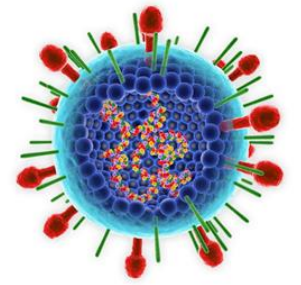
- ▶ **No specific surveillance for pediatric RSV burden till 2023-2024**

→ Decision to build **a broad surveillance network** to better **assess the RSV pediatric burden of disease in Belgium (first in hospitals setting)** and to **monitor impact of preventive tools** once implemented

- ❖ **Adaptation of SARI hospital surveillance** to better catch pediatric RSV cases (since Nov 23)
- ❖ **Creation of the RSVPED study**, run in parallel to SARI survey

- ▶ Both systems in parallel **during 3 seasons**
 - 2023-2024 (pre-prevention period);
 - 2024-2025 (first year of prevention implementation, one tool reimbursed),
 - Future 2025-2026 (second year of prevention, two tools reimbursed)

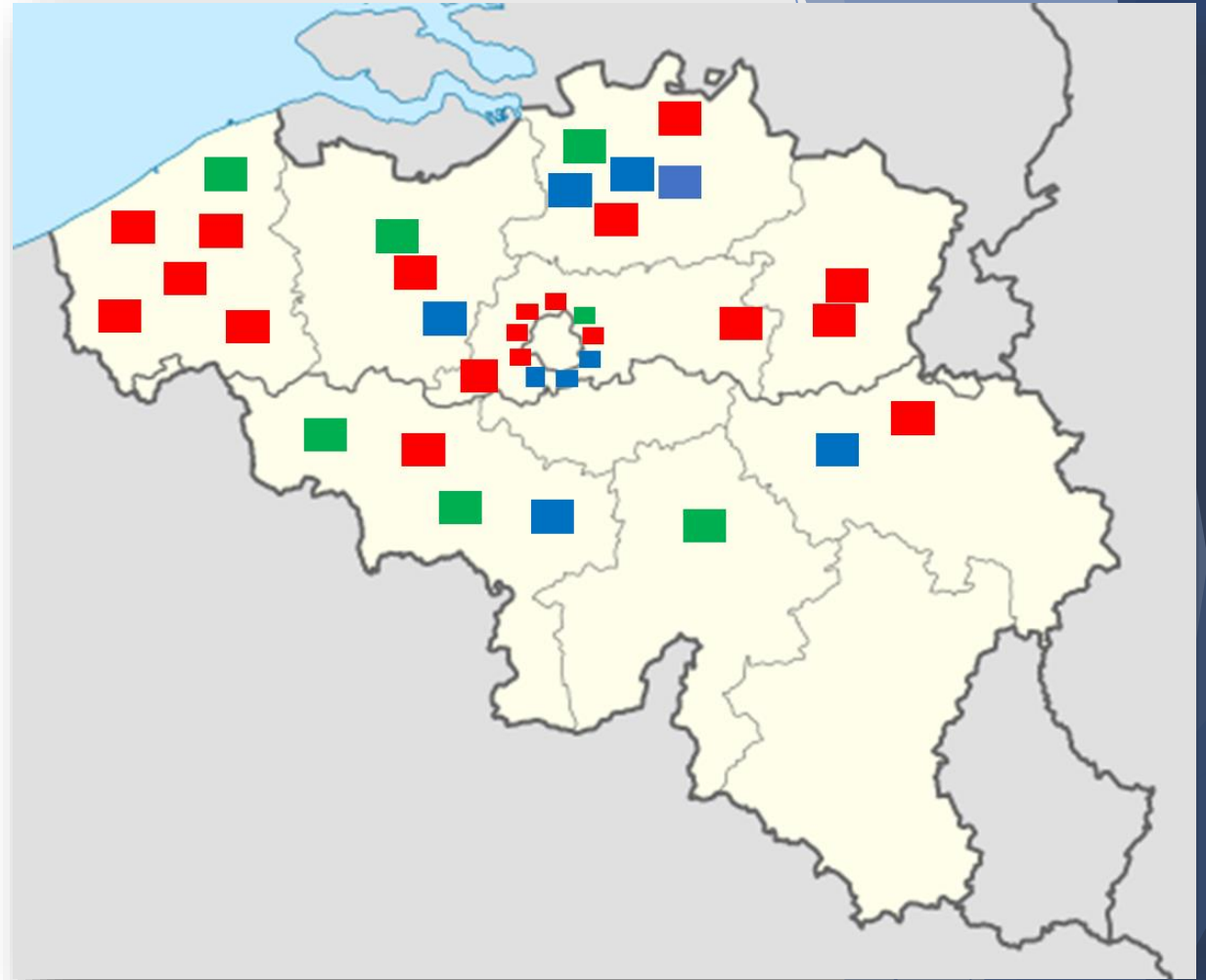
Method



- ▶ New **common case definition** for pediatric severe acute respiratory infections
- ▶ **Questionnaire** adapted to pediatric features, **almost similar between both studies**
- ▶ **SARI surveillance (active since 2013)**
 - Inclusion of all cases hospitalized for severe acute respiratory infections matching the clinical case definition
 - Sample sent to the National Reference Center, tested for 18 respiratory viruses (PCR)
 - Weekly reporting (bulletin of acute respiratory infections)
- ▶ **RSVPED study**
 - Active recruitment of Belgian centers outside the SARI network
 - Inclusion of **all proven RSV** cases **0-4 years old** meeting the case definition and hospitalized during the enlarged epidemic period (October to March)
 - Retrospective **clinical data collection** by local team/data nurse
 - Preliminary **aggregated data** obtained, mostly from the **3 first months of the past season**
- ▶ **Pooled analysis after each season and specific analyses leveraging assets of each study**

34 participating sites all over the country

- ▶ **RSVPED : 27 centers**
 - 19 centers from which incidence data from both seasons are available (paired data)
 - 8 centers partial data collection
 - And many more coming...!!
- ▶ **SARI : 7 additional centers**



UZ Brussels, UZ Gent, AZ Sint Jan Brugge, Centre Chwapi Tournai, CHU Namur-Godinnes, Grand Hôpital de Charleroi, ZAS ziekenhuizen, CHU St Pierre, CHU Tivoli, AZ Klina, AZ Turnhout, AZ Delta Roeselare, Jessa Ziekenhuis, Clinique St Jean, AZ St Maria Halle, AZ Jan Palfijn Gent, AZ Groeninge, Jan Yperman, Hôpital Delta Chirec, Hôpital St Anne-St Remy Chirec, Hôpital de Braine-l'alleul, HUDERF, OLV Wareghem, CHC Mont Leggia, AZ Diest, Imelda Ziekenhuis, Hôpital Iris Sud Ixelles, Vitaz, Cliniques de L'Europe, Cliniques Universitaires St Luc, Universiteit Ziekenhuis Antwerp, CHU/Citadelle Liège, AZ Blasius, CHU Charleroi

RSV: high pediatric burden of disease season 2023-2024

▶ RSV PED

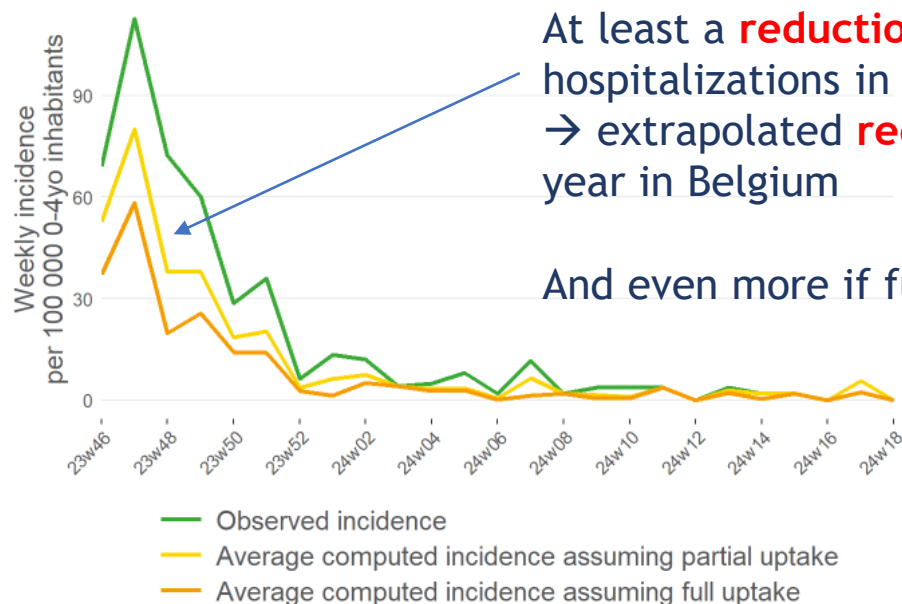
- ▶ **3175 cases** included from **22 centers** (even under estimated!)
- ▶ 53.3% < 6 months old, 90% < 2 years old
- ▶ Mean length of stay: 5.4 days
- ▶ Use of respiratory support in 61% of the cohort (67% in infants <6m)
- ▶ Antibiotics consumption in 34% of children (26% in infants <6months)
- ▶ 217 ICU admissions recorded from 5 PICU tertiary centers



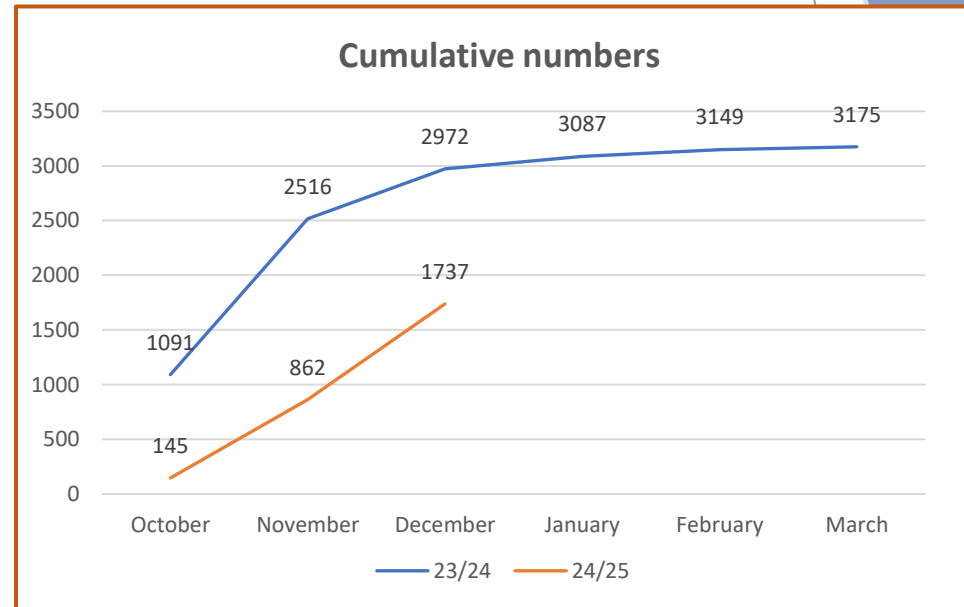
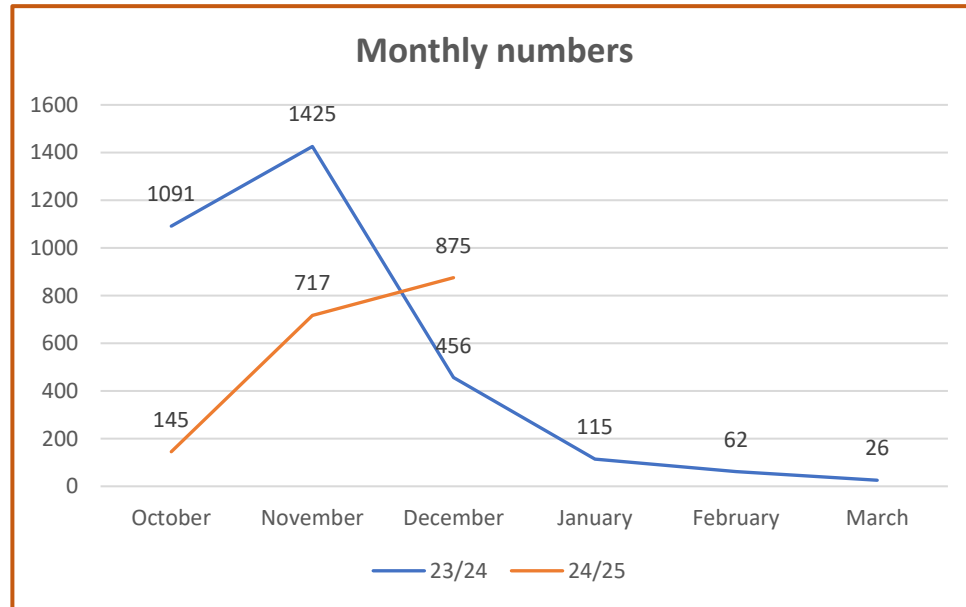
How many hospitalizations could hypothetically be prevented in 2023-2024 if Nirsevimab had been distributed as in the following year? (both studies)

Parameters

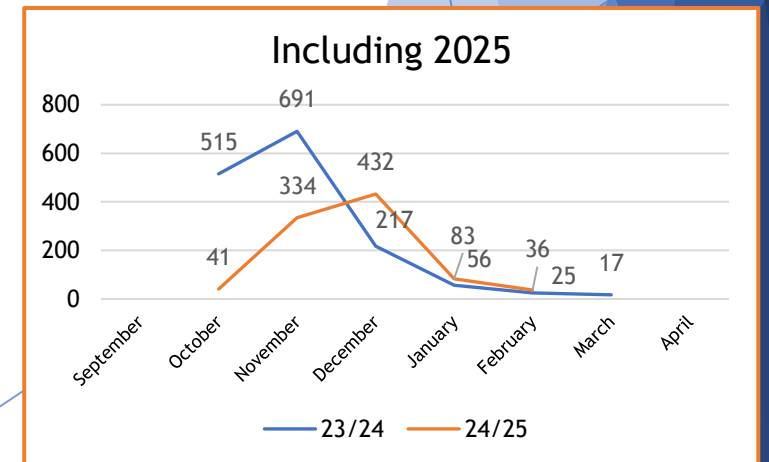
- **Proportion of admitted children** that would have been **eligible for immunization** among all hospitalized cases (before prevention available)
66% (newborns 29%; catch up 37%).
- **Percentage of uptake** of the preventive tool: **85% maternity, 50% catch up**
- **Efficacy of the preventive tool** to prevent hospitalization (literature data) : **88%**



Number of children 0-4 y old hospitalized for RSV in 2024-2025 (first year prevention) compared to 2023-2024 (RSVPED)

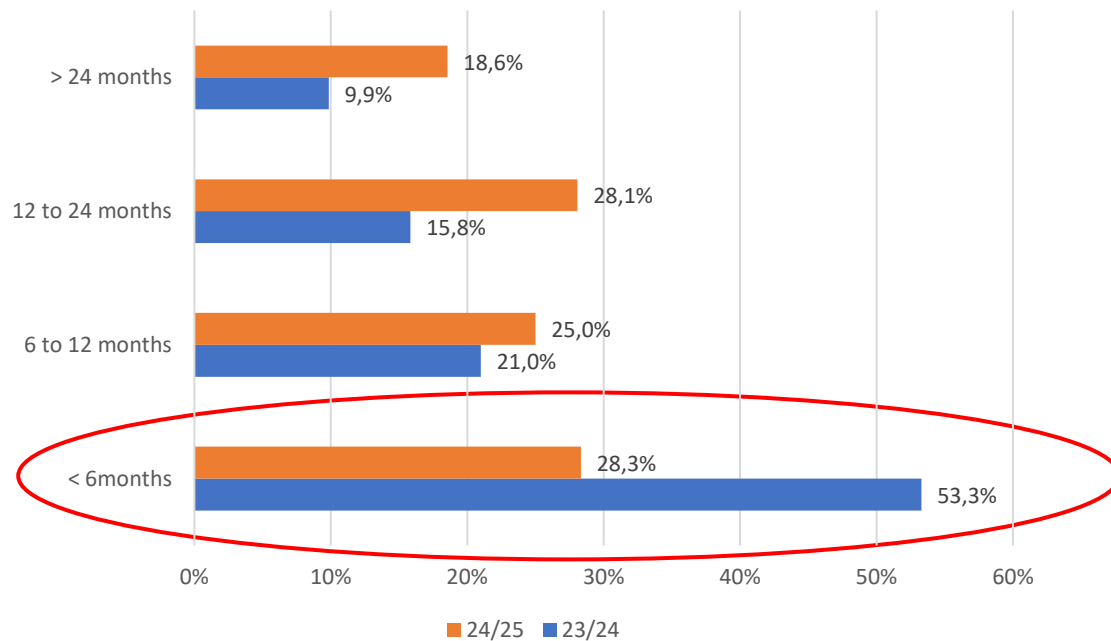


- ❖ Between October and December: **much less hospitalized cases** this season compared to the previous one
- ❖ **Shift in seasonality: previous season started earlier** than in the pre-pandemic period (start in September in 2023, October in 2024)
- ❖ Estimated **reduction of 1750 hospitalizations in the month of the peak**

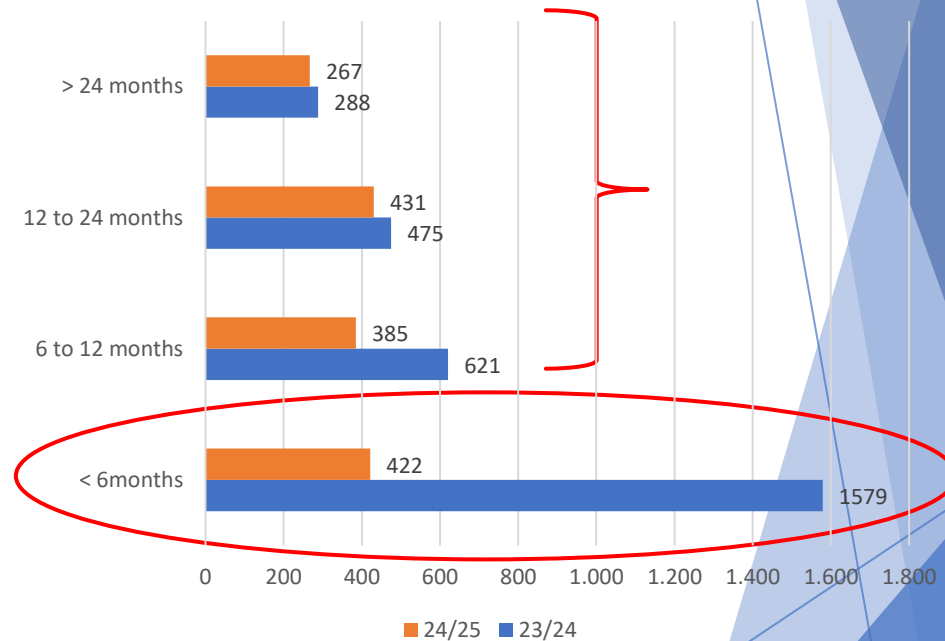


Age distribution among RSV pediatric hospitalized cases during both seasons (RSVPED and SARI)

RSVPed + SARI

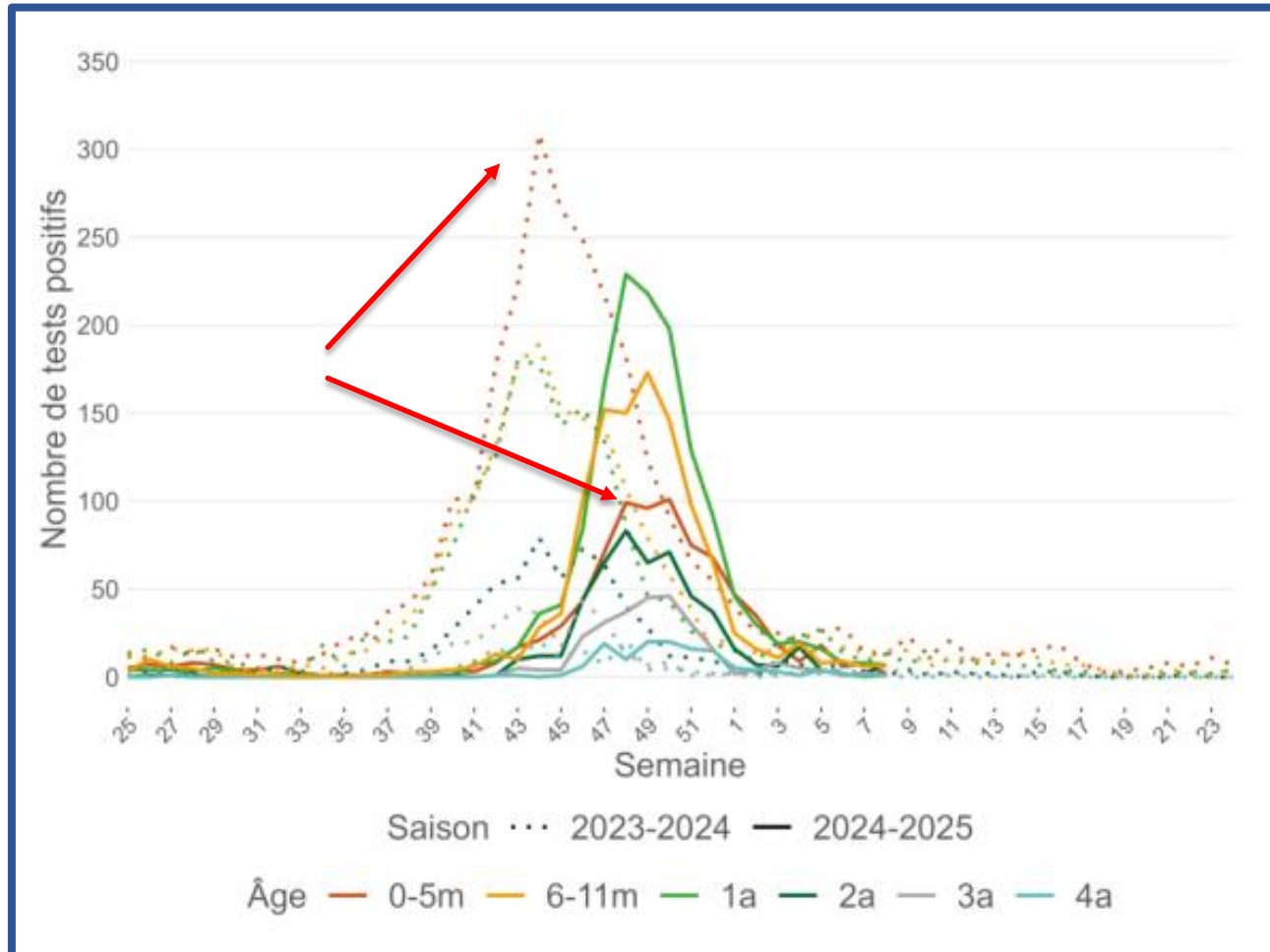


Absolute numbers (RSVPED)



Decrease of burden in <6 months in 2024-2025 (proportionally less represented)
Concordant results between both studies
No changes in other age groups (no age shift)

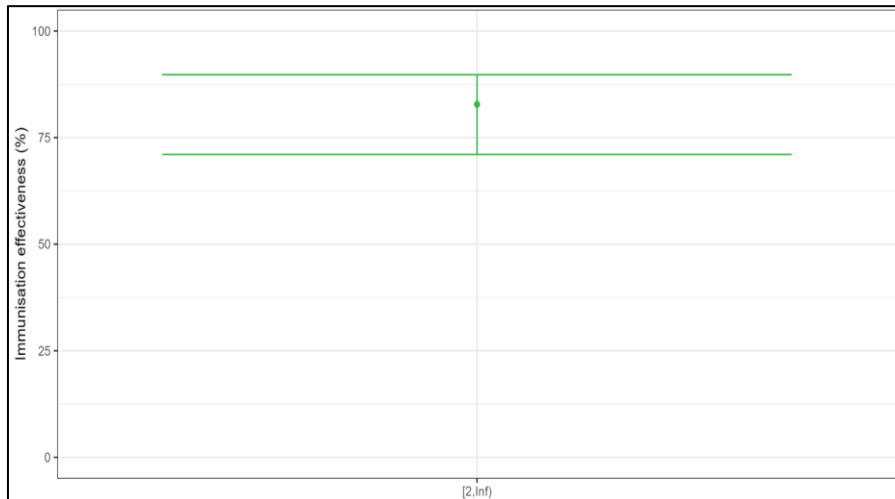
Incidence pattern of ambulatory RSV cases in children 0-4y in 2023-2024 and 2024-2025 (sentinel labs, Sciensano)



Effectiveness of nirsevimab in preventing hospitalization (SARI, 2024-2025)

Test-negative design (common in vaccine studies):

- Compares nirsevimab-eligible SARI patients hospitalized because of RSV (**cases**) to those hospitalized for SARI but tested negative for RSV (**controls**).
- Adjusting for **confounders** : comorbidities, sex, sampling week, age in months
- **Excluding** nosocomial infections, children who received palivizumab or whose mother received Abrysvo
- Immunized children represented around 20-25% of RSV hospitalizations of eligible children



Immunization effectiveness = 83%
(logistic regression)

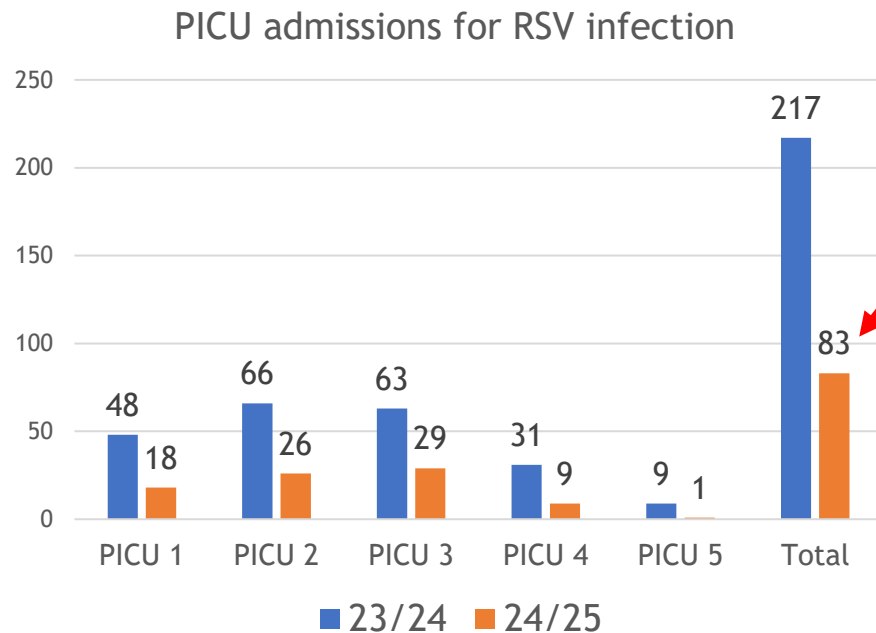
	RSV-negative SARI hosp	RSV-positive SARI hosp
Not immunized	111	170
Immunized	107	36

- No decrease of effectiveness with time, until 90 days post immunization

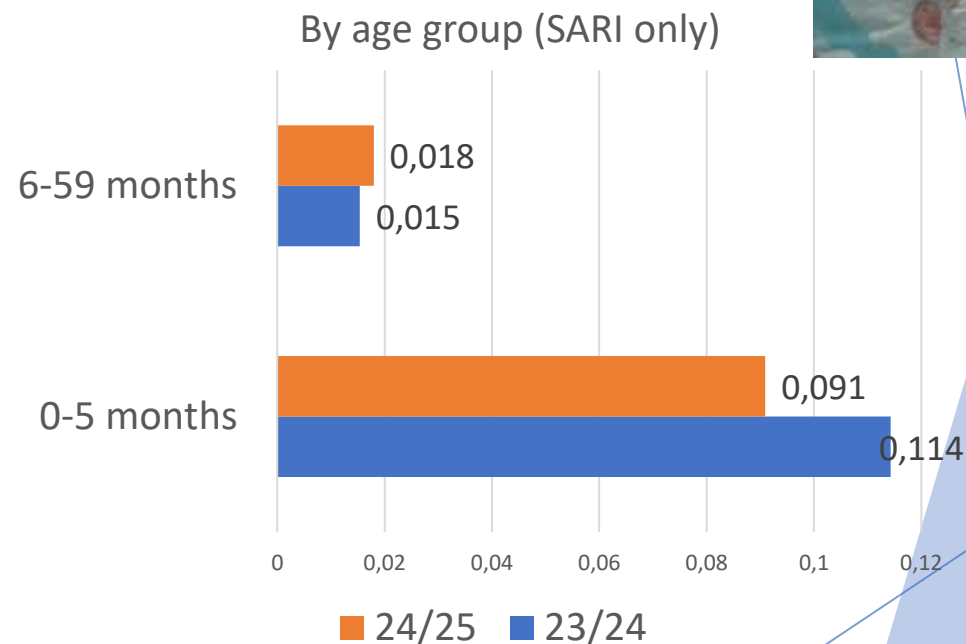
Severity of infections/burden of disease

ICU admissions (both studies)

Decrease of total ICU admissions for RSV (5 main PICU centers, 3 regions)



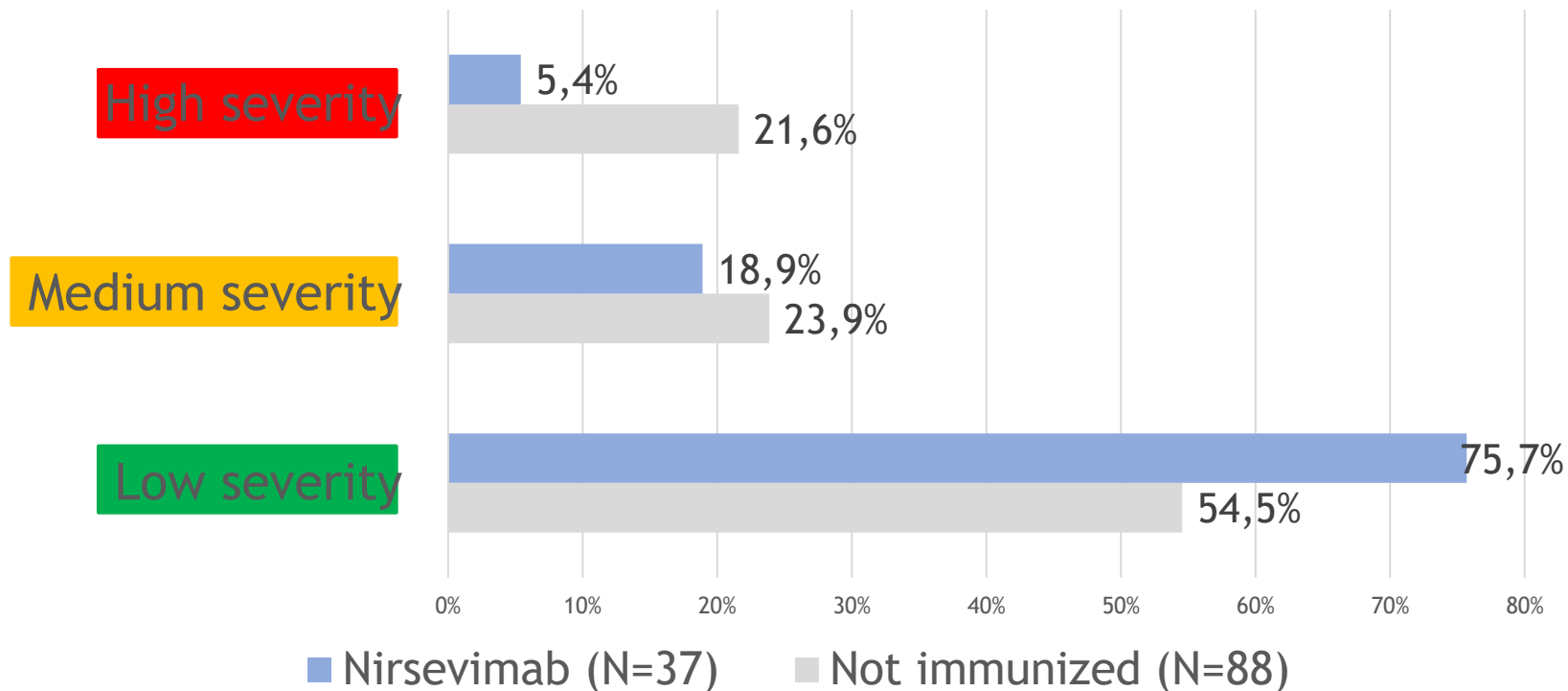
Ratio children admitted to ICU over total hospitalized in non-PICU hospitals



The burden of ICU admissions has decreased...

... and it is not only the effect of the decrease in hospitalizations

Severity of infections in infants eligible for Nirsevimab according to immunization status (SARI, 2024-2025)



3 grades of severity
according to use of
respiratory support

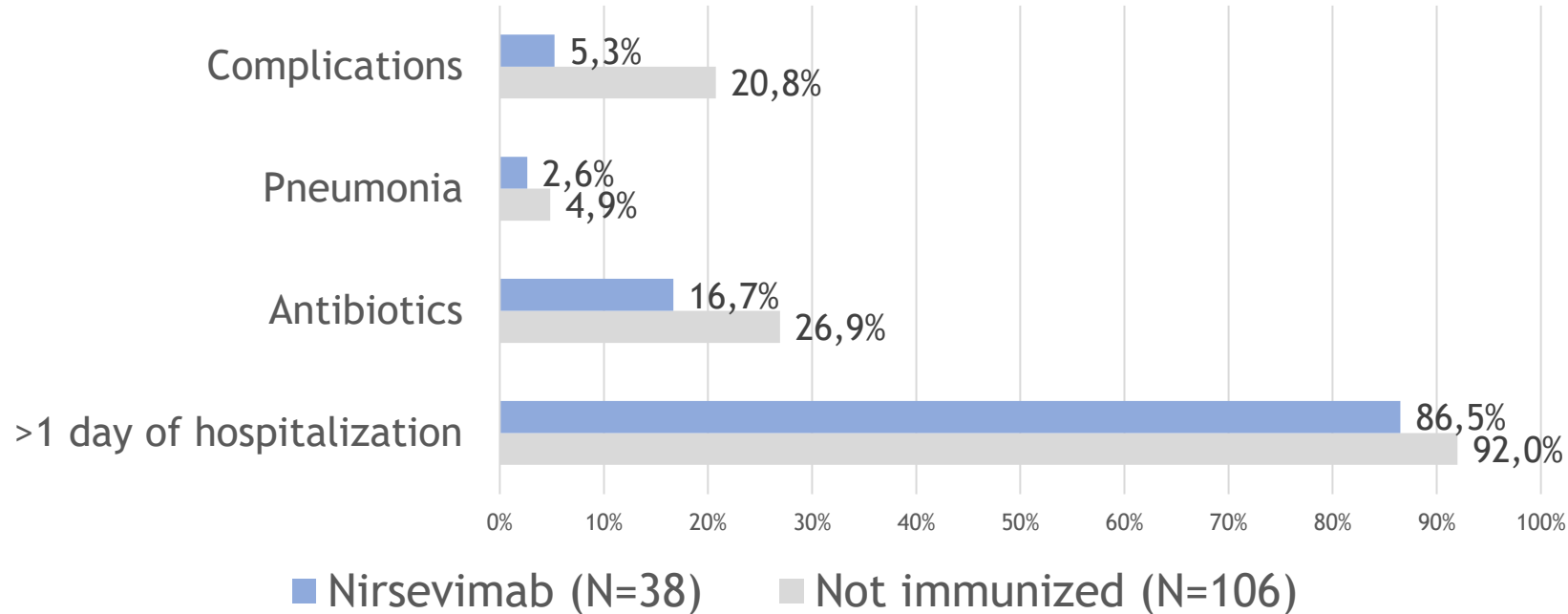
(3) = highest severity:
invasive ventilation or ICU
admission

(2) = mid-severity: non-
invasive ventilation or oxygen
goggles

(1) = low-severity: none

Infants immunized although hospitalized showed less severe infections!

Severity of infections in infants eligible for Nirsevimab according to immunization status (SARI, 2024-2025).



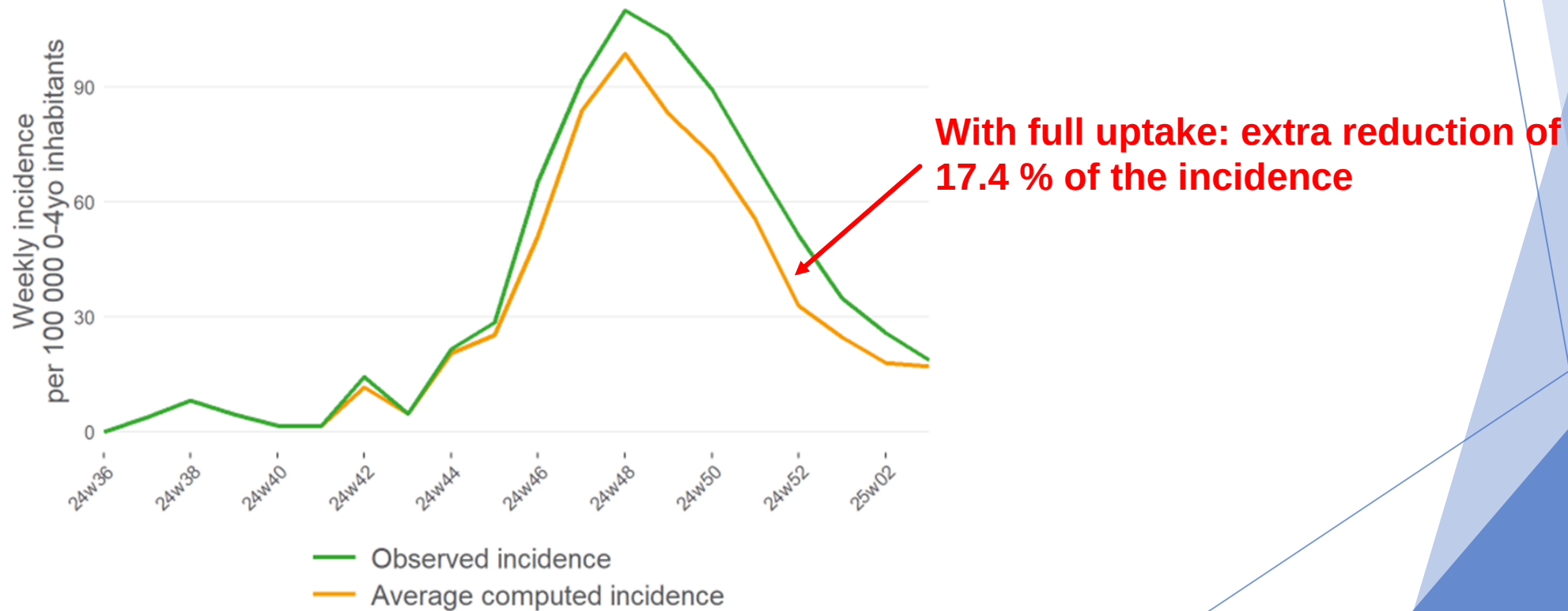
To be confirmed with RSVPed data, with addition of other severity indicators

e.g. need for nutritional support, rate of neurological complications, total nb of transfers to ICU, details of ICU stays...

+ Severity in older age groups (> 6m)

If all eligible patients would have received Nirsevimab (100% uptake) this season (Oct 2024-Janv 2025, SARI data)

Non-immunized children represent around 75% of RSV hospitalizations of eligible children



Take home messages

- ▶ **Reduction of RSV pediatric hospitalizations** in 2024-2025 (first season with RSV prevention) compared to 2023-2024
- ▶ **Reduction of RSV PICU burden and severity**
- ▶ **Highest impact** seen among the **<6 months old** (in terms of number of cases and severity)
- ▶ **Immunization effectiveness > 80%**
- ▶ **We can do better with bigger uptake**

- ▶ Perspectives:
 - ▶ **In-depth analysis of RSV PED data and comparison between both seasons** of pattern of RSV infection, antibiotics consumption, coinfections rate and complications
 - ▶ **Cost effectiveness analysis from KCE ongoing**
 - ▶ **Changes in upcoming seasons:** new tools (Abrysvo, then clesrovimab, vaccines for older children), changes in reimbursement criteria and uptake

Acknowledgements

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- ▶ The Care Foundation Chirec



